

Final Final Project Writeup, STAT 488 Spring 2024

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May 2, 2024

Paper Description/Context

Many crises that arise in humanity follow an archetype of sorts: solutions found for problems often come with scientific advance. One of the most salient and recent crises to have occurred is the COVID-19 pandemic. The disease is emerged in Wuhan, China and was declared a pandemic on March 11, 2020. Research exists on SARS-CoV-1, a related virus, but this is relatively sparse compared to the research required at the time to mount a response to SARS-CoV-2.

One of the main differences between the endemic caused by SARS-CoV-1 and the COVID-19 pandemic is the renewed interest in machine learning - while the theory for many machine learning methods have been around for a long time, datasets were too small and the calculations required hardware that simply did not exist at the time. With the dawn of cloud computing, big data, and advances in parallel computing, machine learning as a field experienced a renaissance, and became an important tool in determining the current state of the pandemic, predicting how the pandemic may change over time, and planning interventions to help slow the spread the infection and improve patient outcomes of those infected with the disease.

In a paper published 2021, researchers Chaudbary and Singh titled [Community Detection Using Unsupervised Machine Learning Techniques on COVID-19 Dataset](#), the authors hoped to use the unsupervised machine learning method, K-means clustering, coupled with principal component analysis dimension reduction in order to create clusters (communities) of countries requiring special attention when implementing a response to the pandemic. Previous studies on the pandemic were usually traditional statistical analyses on exploratory and explanatory bivariate analysis (at the time of the publication) and the topic of study often focused on the end user (patient), such as metrics on biochemistry or pathology.

The authors of this paper wanted to create a more macroscopic analysis of the epidemiology of COVID-19, but the results of this paper were not intended to be clinical, but a proof of concept for the use of

clustering algorithms to find in this case countries that required additional help for mounting a response to the pandemic, but for big data in healthcare and epidemiology in general.

Data Description

The dataset being analyzed was scraped from the [Johns Hopkins Coronavirus Resource Center](#) via a [scraping tool](#) obtained from GitHub. The scraping tool takes information from the resource center and updates weekly with new data - some of the data being accumulatory (a total that increases over the course of the pandemic), others being updated weekly (such as percent change during the week). The data consists of $n = 197$ countries across 15 variables, where $p = 13$ are continuous and will be used for the bulk of the analysis. Further information about the individual variables can be found in Appendix B.1.

The dataset used in this paper contained the data collected on August 15th, 2020, and scraping began January 22, 2020. The data that was updated weekly is represents those variables collected on August 15th 2020, and the accumulating variables accumulated from January 22, 2020 to August 15th, 2020. The scraper, however, stopped collecting data once Johns Hopkins ended data collection on its surveillance feed on march 10, 2023, and so the dataset used in this analysis covers until March 10th, 2023. There is no way to obtain the historical data (unless the authors are contacted directly), as the scraper replaced old data without archiving.

The variables were renamed for ease of reference (Appendix B.2), and cleaned so that NaN's (which were not found) or infinite values (which were set to 0) were no longer in the dataset.

Project Aims/Goal (C)

The authors had one primary goal in this paper: to show K-means clustering with PCA as a proof of concept for clustering countries into different groups based on the metrics collected by the scraper. This analysis seeks to further this goal by using the PCA to find which variables are most important in terms of explaining the variance in the dataset and elucidate reasons why the countries were clustered.

Description of Analysis Methods Used (D)

Results and Interpretations (E)

Final Conclusions and Further Analysis (F)